Riddhi Agnihotri

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Education

Vellore Institute of Technology, Bhopal **2023–2027**
Bachelor of Technology (CSE – Specialization in Gaming Technology) **CGPA: 8.86**

St. Stephen's School, Kolkata **2022–2023**
Higher Secondary School Certificate **77.83%**

St. Stephen's School, Kolkata **2020–2021**
Secondary School Certificate **88%**

Projects

Celestia Probe – Agentic AI Space Intelligence System *LangGraph, LangChain, Chroma, FastAPI*

- **Multi-Agent Architecture:** Built a LangGraph-based agentic pipeline for query planning, archive validation, knowledge acquisition, retrieval, and grounded response generation.
- **Dynamic RAG System:** Developed a self-expanding Chroma vector knowledge archive that identifies knowledge gaps and dynamically acquires content from NASA and Wikipedia.
- **Retrieval Pipeline:** Implemented document chunking, embeddings, vector similarity search, and context-aware retrieval using LangChain and Chroma.
- **Real-Time AI Streaming:** Integrated FastAPI and Server-Sent Events to stream agent execution states and generated responses to the frontend in real time.
- **Immersive AI Interface:** Designed and developed a custom CRT-inspired HTML, CSS, and JavaScript terminal interface for interactive scientific exploration.

YOLOv7-Based Threat Detection System *PyTorch, YOLOv7, Computer Vision*

- **Computer Vision Pipeline:** Built a real-time object detection system using YOLOv7 and PyTorch to identify potential threats in surveillance imagery.
- **Multi-Class Detection:** Trained a custom detection model to recognize 9 weapon categories across surveillance scenarios.
- **Model Optimization:** Reduced inference input resolution to 416×416 to improve detection speed and optimize execution on limited GPU hardware.
- **Custom Training Pipeline:** Configured YOLO-format datasets, annotations, training parameters, confidence thresholds, and model evaluation workflows.
- **Detection Visualization:** Implemented bounding-box and class-label outputs for real-time monitoring and model result interpretation.

FOSSEE (IIT Bombay) Workshop Booking Platform Redesign *Figma, React, UX Research*

- **UI/UX Redesign:** Reimagined and rebuilt the workshop platform interface to improve usability, visual hierarchy, and user flow.
- **Problem-Driven Design:** Diagnosed and resolved critical UX issues including weak CTAs, unclear navigation, and inconsistent layouts.
- **Improved Data Presentation:** Enhanced readability of tables, filters, and dashboard sections with structured layouts and feedback states.
- **UX Transformation:** Transformed a fragmented, inconsistent interface into a structured, intuitive, and user-guided experience.

Technical Skills

Languages:	Python, Java, C++, JavaScript, SQL
AI/ML & Agentic Systems:	LangGraph, LangChain, RAG, Chroma, YOLOv7, PyTorch, scikit-learn, Vector Embeddings
Backend & APIs:	FastAPI, REST APIs, Server-Sent Events (SSE), API Integration
Frontend:	React, HTML, CSS, JavaScript
Databases & Storage:	MySQL, SQL, Chroma Vector Database
Tools & Platforms:	Git, GitHub, AWS, Unity, Figma, Tableau

Certifications & Achievements

- **Top 5% Nationwide – NPTEL Human-Computer Interaction (IIIT Delhi):** Scored 95/100.
- **Microsoft DP-900: Azure Data Fundamentals:** Scored 970/1000.
- **3 Month Remote Paid Internship:** Administrative Assistant at Orange Tribe (Branding and Packaging Studio, Noida)
- **Academic Excellence:** Achieved S grades in 7 subjects, including Applied Linear Algebra, Fundamentals of AI and ML, Game Theory, Game Programming using Unity, and Introduction to AR/VR/XR.